

we can examine what occurs in their brains as they make decisions. Technology like fMRI (functional magnetic resonance imaging), EEG (electroencephalogram), eye tracking, blood chemistry analysis, and galvanic skin response all aid in interoception (our ability to identify signals inside the body).

Through a series of clever experiments, Camerer has discovered that some of our decision-making takes place in milliseconds—even before we have any conscious awareness. In response to a threatening image that lasts just 30 milliseconds—the length of a single frame in a movie (films use 24 frames per second)—fMRIs spot increased activity in the amygdala. Even before we are consciously aware of a threat, our brains have already decided upon a course of action.

The brain's frontal cortex distinguishes modern human intellect from that of our ancestors. The fast-responding parts of our deep and midbrain are ancient. These are where we find faults in our wetware.

To paraphrase Daniel Kahneman's book title, our lizard brains think *really fast* and slow.



Dopaminergic neurons, which make up less than 1% of the brain's neurons, are the primary source of dopamine in our nervous system. When we engage in activities like sex, drinking, eating, gambling, etc., these neurotransmitters flood the brain's reward circuitry with dopamine. Feelings of pleasure, satisfaction, and motivation are the result.

Psychiatrist Anna Lembke studies how some indulgences increase dopamine production above a baseline level.^{[368](#)} Chocolate increases dopamine by 50%; sex causes a 100% increase; nicotine a 150% surge.^{[369](#)} For smokers, this implies the cigarette after sex is more pleasurable than the sex itself.^{[370](#)}

Our decision-making process includes much that we are not consciously aware of. If we do not know why we make specific decisions, how can we